Regression

Multiple Linear Regression

Linear Regression Assumptions

- **Linearity.** X and Y should have a linear relationship,

$$Y = \beta_0 + \beta_1 X + \epsilon.$$

  Check using *scatter plots.*
- **Residual Independence.** The residuals should generally be random. Check using:
  - *Time dependence* $\Rightarrow$ scatter plot of residuals over time;
  - *Spatial dependence* $\Rightarrow$ map...
- **Constant variance of residuals.** The residuals have a mean of 0 and should have a constant variance.
  Check using a *fited(predicted) vs residual plot*.
- **Normality of residuals.** We assume that the residuals are normally distributed.
  Check using a *quantile-quantile (QQ) plot*.

Multiple linear regression

- Model: $Y = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p + \epsilon$
- Predictor data:

$$\begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

- Fitted model (predictions):

$$\widehat{y}_i = \widehat{\beta}_0 + \sum_{j=1}^{p} \widehat{\beta}_j x_{ij}, \quad i = 1, ..., n.$$

Regression coefficients

- Interpreting regression coefficients.
  $Y = \beta_0 + \ldots + \beta_j X_j + \ldots + \beta_p X_p + \epsilon$. (Hyphothetically) a unit change in X_j is associated with a β_j change in Y, while all the other variables stay fixed.

- Correlations among predictors cause problems: *how can other variables stay fixed if they are correlated with X_j?*

- Estimation. Estimate the coefficients by minimizing the Residual Sum of Squares

$$RSS = \sum_{i=1}^{n} \left(y_i - \sum_{i=1}^{n} \left(\widehat{\beta}_0 + \sum_{j=1}^{p} \widehat{\beta}_j x_{ij} \right) \right)^2$$

Matrix form

Model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

$$\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}.$$

Prediction:

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}.$$

Matrix form

- Residual Sum of Squares:

$$\begin{aligned} RSS(\widehat{\beta}) &= ||\mathbf{y} - \mathbf{X}\widehat{\beta}||^2 \\ &= (\mathbf{y} - \mathbf{X}\widehat{\beta})^T(\mathbf{y} - \mathbf{X}\widehat{\beta}) \\ &= \widehat{\beta}^T(\mathbf{X}^T\mathbf{X})\widehat{\beta} - 2\mathbf{y}^T\mathbf{X}\widehat{\beta} + \mathbf{y}^T\mathbf{y} \end{aligned}$$

- Minimizing the $RSS \Rightarrow$ the optimality conditions:

$$\left(\mathbf{X}^T\mathbf{X}\right)\widehat{\beta} = \mathbf{X}^T\mathbf{y}.$$

- Solution (What is the condition on the design matrix **X**?)

$$\widehat{\beta} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y}.$$

- Residual vector:

$$e = \mathbf{y} - \mathbf{X}\widehat{\beta} = \mathbf{y} - \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y}$$

Hyphothesis Testing: Multiple linear regression

Is at least one predictor useful in predicting the response Y?

- **Hypothesis testing.** This corresponds to testing

  $$H_0 : \beta_1 = \ldots = \beta_p = 0 \text{ (none is useful)}$$

  versus

  $$H_1 : \beta_i \neq 0, \text{ for some } i \in \{1, .., p\} \text{ (at least one is useful).}$$

- To test the null hypothesis, we compute the F-**statistic**, given by

  $$F = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}.$$

  Similar to the one dimensional case, a p-value associated with the F-statistic is computed. The null hypothesis is rejected when the p-value is sufficiently small.